



Cross-country variation in household access to financial services[☆]

Patrick Honohan^{*}

Institute for International Integration Studies, Trinity College, Dublin, Ireland

ARTICLE INFO

Article history:

Received 4 March 2008

Accepted 17 May 2008

Available online 24 May 2008

JEL classification:

G2

O1

Keywords:

Access to finance

Financial exclusion

Microfinance

Worldwide database

ABSTRACT

This paper presents estimates, for more than 160 countries, of the fraction of the adult population using formal financial intermediaries. The estimates are constructed by combining information on account numbers at banks and microfinance institutions (together with banking depth and GDP data) with estimates from household surveys for a smaller set of countries. An illustrative application of the data compares them with information on poverty: there is a correlation, but it is not clearly causal.

© 2008 Elsevier B.V. All rights reserved.

1. Introduction and summary

This paper presents estimates across some 162 countries of the fraction of the adult population using formal (or semi-formal) financial intermediaries, whether through deposit accounts or by borrowing. The estimates are constructed by combining information (assembled by others) about the number of accounts at commercial banks and at microfinance institutions (MFIs) (together with banking depth and GDP data) with estimates from household surveys for a smaller set of countries. As a first application of the data, the hypothesis that more access lowers poverty is given a preliminary test.

The financial sector plays a multidimensional role in the process of development. It mobilizes and concentrates resources for investment and allocates them based on an assessment of risk and return, judging creditworthiness and monitoring performance; it offers risk-reduction and risk-pooling services that have both direct effects on welfare (by providing insulation from shocks) and indirect effects on growth, by making riskier – but potentially high-yield – investments in human and physical capital accessible. While much of the recent literature focuses on the interaction of enterprises, large and small, with financial markets and financial

intermediaries, households are important consumers of financial products, and household behavior influences the scale and asset mix of finance (Honohan, 2008).

Because financial asset holdings are highly concentrated, even more so than nonfinancial assets, the assets of the lower quartiles can be largely ignored in any discussion of national resource mobilization or aggregate financial wealth. Nevertheless, it appears that countries with deeper financial systems have less absolute poverty, even after taking account of mean national income (Beck et al., 2007a; Honohan, 2004a). This suggests that a deep financial system also affects the structure of the economy in indirect ways that are relevant for the inclusiveness of economic development.

As elaborated in Section 2 below, it is possible that poverty and inequality could be strongly influenced by the degree to which low income households have access to savings, risk-pooling and payments services provided the formal financial sector. Indeed, the main focus of current policy concern with regard to household finance in developing countries is the perceived need to increase the access of poor households to basic financial services: deposits, payments, insurance and credit.¹

Relevant policies include regulatory design for microfinance institutions designed to ensure that measures intended to protect consumers against loss do not impose costs so heavy as to deter entry into this low end of the market, which is characterized by low

[☆] Thanks to Thorsten Beck, John Caskey, Jim Davies and the referees for helpful suggestions. This paper was originally submitted to Professor Giorgio Szego on 22 November 2007 and subsequently a revised version was submitted through EES.

^{*} Tel.: +353 1 896 3195; fax: +353 1 896 3939.

E-mail address: phonohan@tcd.ie.

¹ To be sure, credit can be a two-edged sword, and over-indebtedness can be a crippling burden for households, a point which needs to be borne in mind in designing consumer-protection policies for household finance (Honohan, 2004b).

margins despite the potentially high volume. The same applies to the design of measures intended to protect the integrity of the financial system against money laundering or its use to finance terrorism; the application of these anti-money laundering and countering the financing of terrorism (AML/CFT) measures to providers of financial services at the small scale needs to be proportionate to the risks involved and adapted to the capacity of the providers to comply.

Yet the body of research on what impact microfinance has had on poverty is surprisingly inconclusive. Although beneficiaries of microfinance schemes are vocal in their praise and gratitude, a comprehensive assessment needs to consider displacement effects and endogeneity of financial access. Only in a few cases so far has it been possible to devise convincing ways of adjusting for, or excluding, these complicating factors. On balance, most observers regard microfinance interventions as poverty-reducing (cf. [Armendariz de Aghion and Morduch, 2005](#)), while continuing to call for further analysis of methodologies for increasing cost-effectiveness and sustainability of these initiatives. After all, most microfinance schemes continue to benefit from external subsidies.

An alternative approach to detecting a direct influence of financial access on poverty and inequality requires the assembly of cross-country data. Quantifying household access on a cross-country basis is the focus of Section 3. This describes and implements a method for combining data on client and account numbers at microfinance institutions and banks with the results of household surveys in a number of countries and macroeconomic data to generate a composite estimate of the fraction of adults using² the services of formal³ financial intermediaries. The data clearly shows that this percentage is small in most developing countries. In contrast, access percentages in the 80s and 90s are recorded for households in advanced economies (where the relevant policy issues are therefore best discussed in the language of exclusion rather than of access). The new series is regressed on a number of structural features of national economies: among the strong correlates of household financial access are mobile phone penetration and the quality of institutions.

The resulting data are compared with information on poverty in Section 4. Although there is a bivariate correlation, multivariate analysis finds no convincing evidence that access is causally related to a lower poverty headcount.

2. How could access to a bank account help reduce poverty?

Casting our eyes to the bottom of the pyramid, we need to remind ourselves that, for lower income groups, the relevant question concerning their financial assets is not related to how much financial assets they choose to hold, and even less to how much they might choose to borrow from the financial system. These households encounter barriers to accessing financial services at all.⁴ This has led to a growing debate around the issue of financial exclusion in the advanced economies, where it is noted that a small group – perhaps 10% of the adult population – which suffer deprivation along multiple dimensions – do not have access to financial services such as a transactions account, or even a savings account, much less a loan from a formal intermediary or an insurance policy.

² There is an obvious and important conceptual distinction between access and usage, and surveys often explore this (for further discussion, see [Honohan, 2004b](#)). In the present paper we treat the terms synonymously, which may not be too bad an approximation when equating access to the use of any financial product.

³ The discussion is generally confined to the formal (and semi-formal) financial system and as such excludes borrowing from informal lenders and participation in, for example, informal rotating savings and credit schemes. The term “semi-formal” is intended to include microfinance institutions which might not be subject to regulation as financial intermediaries.

⁴ Including transactions services – an important dimension for such households which can be associated with the holding of balances in transactions accounts, but which will not be further discussed in this paper.

In some cases product features, such as a high minimum cover for an insurance product, or heavy penalties for unauthorized overdrafts (hard for poor people to avoid if they are using checking accounts), or having a fixed address as a pre-requisite to open an account, represent material obstacles to use of such products. In other cases customers may be screened-out because of risk characteristics. Or rationalization of branches and service points may result in many poor customers having too far to travel to a branch to make an account worthwhile. Given the increasing extent to which full participation in economic life in the advanced economies depends on having an account at a financial intermediary, and given the material extra costs often imposed on non-account holders, several countries have adopted policies in recent years to reduce financial exclusion ([Carbó et al., 2005](#); [Kempson et al., 2000](#)).

In low and middle-income countries, though, ‘exclusion’ is normal for the bulk of the population, hence it becomes more natural to speak of broadening access to financial services rather than elimination of exclusion as the immediate goal. The exploding microfinance movement is driven by this motivation ([Armendariz de Aghion and Morduch, 2005](#); [Honohan, 2004b](#); [Robinson, 2001](#)). Microfinance pioneers have emphasized the very high rates of return that can be earned by the poor and the near-poor, especially in urban and peri-urban settings, resulting in a high demand for borrowings even at high interest rates. Some microfinance institutions work on a credit-only basis, funding themselves from charitable donors and other sources; some employ forced savings elements to the loan scheme. The modern trend is to emphasize deposits as well as loans as a key tool for efficient financial management, whether they represent precautionary savings, or a means of accumulating capital. Transactions accounts are also important for receiving and making domestic and international remittances between family members. And microinsurance is beginning to be a significant element in the microfinance movement.

Does access to financial services through deposit or loan accounts reduce poverty? Numerous anecdotes illustrate paths to relative prosperity being paved by such financial services. However, there is typically a large element of selection bias, both at the level of the individual client (with the more energetic likely to experience growing loan and deposit balances) and at the level of the village selected for establishment of a microfinance institution (sometimes this is a negative bias, as charitable sponsors seek out the more deprived villages). Furthermore, there can be displacement effects, with non-beneficiaries of microfinance suffering in local markets from the beneficiaries. Despite an extensive quasi-promotional literature, detailed microeconomic analyses are, for these technical reasons, surprisingly non-committal about whether direct financial access has a major effect in reducing poverty ([Honohan, 2004b](#)).

What can be said at the macro level? Addressing this question evidently requires a cross-country data series on access. There are many dimensions to financial access, but to simplify matters, given the shortage of data, it is preferable to concentrate on one summary indicator. The most widely accepted indicator is the percentage of adults that have any type of account (including all types of transactions, savings or loan accounts) at a bank or other formal financial intermediary. How many people in developing countries have access to a bank or other formal financial intermediary account? This empirical question is the subject of vigorous current investigation as described in the next section.

3. Measuring household access to financial services

This section discusses some recent contributions to measuring access of households to various forms of intermediary accounts, and proposes a new composite cross-country indicator of the per-

centage of the adult population with access to some intermediary account. The new series has the advantage of wide coverage. However, it does rely to an extent on some imputations which might be thought to lack a very robust justification.

3.1. Recent contributions

In recent contributions, two basic approaches have been used to get to grips with the problem of measuring overall financial access (Honohan, 2005). One is to go directly to households and survey their usage, whether as part of a wider household survey or a dedicated financial access survey. The other is to make inferences from data in the number of intermediary accounts. Here we propose a method of using overlaps between the country coverage of the different approaches to arrive at access estimates for 162 countries – or about three times the number of countries for which there is direct household survey data.

The first method of measuring access, namely the use of sample surveys of households or individuals, is the most satisfactory from a conceptual point of view. The surveys that have produced data of this type include general household surveys such as the living standards measurement surveys (LSMS) (which can include a savings module asking questions about the holding of savings) and specialized financial access surveys, such as the Finscope Trust series which began in Southern Africa and are currently being rolled out in several other African countries (www.finscope.co.za).⁵ The series of Eurobarometer surveys in the European Union also ask such questions (cf. European Commission, 2005). Unfortunately the number of countries covered by household survey data remains modest. Indeed, about half of the countries for which direct survey data is available are in the European Union. Our data includes household survey-based data for just eight African countries, four Asian countries, and seven from Latin America. There are indications of problems with the reliability some of the household survey data collected.⁶

The second method is to collect data on the number of accounts maintained at the financial institutions. Data on accounts at *micro-finance* institutions has been assembled for 148 developing countries by Christen et al. (2004) of the consultative group to assist the poor (CGAP), building on earlier compilations for more limited groupings. They include not only specialized microfinance institutions, but also savings and credit cooperatives, credit unions and other socially-oriented intermediaries (including some microfinance-oriented commercial banks) and they term these ‘alternative’ intermediaries.⁷ More recently, Peachey and Roe (2006) have augmented the CGAP database with figures for a number of *additional savings banks* (members of the World Savings Banks Institute), which had not been included by Christen et al. We describe this augmented set as the “WSBI database”. The additions are quantitatively important. Many of the institutions added by Peachey and Roe, most of them postal savings banks, have very wide outreach by national

Table 1

Relating survey-based access percentages and information on account numbers

Regression: Dep. Var.	1.1		1.2	
	No. of bank accounts per 1000 pop		Average bank deposit size % GDP	
Explanatory var.	Coefficient	t-Stat	Coefficient	t-Stat
Constant	156.7	2.5	1356	2.8
Alternative accts per 1000 pop	1.29	5.9		
GDP per cap			0.280	9.4
Outliers omitted?	ROM		No	
R-squared/NOBS	0.762	13	0.822	21
Adjusted R-squared	0.740		0.813	
S.E. of regression	178.3		1735	
	1.3a		1.3b	
Percentage with access (household survey data)				
Constant	–188.2	9.5	–186.1	9.5
Log (bank accts per 1000 pop)	18.2	7.3	18.8	8.0
Log (average deposit size)	14.8	4.7	14.0	4.9
Outliers omitted?	No		No	
R-squared/ NOBS	0.869	30	0.931	17
Adjusted R-squared	0.859		0.921	
S.E. of regression	11.5		10.0	

** and * indicates significance at the 1% and 5% levels, respectively.

** indicates significance at the 1% level.

Data sources: Household survey data from the surveys listed in Claessens (2006) and European Commission (2005); Alternative accts is the number of accounts at “alternative financial institutions” from Peachey and Roe (2006) as amended (see Footnotes 7 and 8 above); data on bank account numbers and average size from Beck et al. (2007b).

Regression 1.3a employs projected values of the explanatory variables based on Regressions 1.1 and 1.2 above. Regression 1.3b is restricted to countries for which the data is available.

Omitted observations: regression 1.1 omits the outlier Romania.

standards: in 27 countries they represent more than 50% of the number of accounts recorded in their database. Furthermore, the additional banks include two very large ones (the Industrial and Commercial Bank, ICBC, of China and Sberbank, Russia, with 0.43 billion and 0.23 billion accounts respectively); these two banks alone account for the fact that the augmented dataset contains more than twice as many accounts – 1.46 billion instead of 0.67 billion – thereby sharply revising aggregate access estimates for the World.⁸

A parallel exercise for *banks* was conducted mainly through a survey of banking regulators (Beck et al., 2007b). Data on the number of deposits at 1999–2000 was obtained for 45 countries – 35 of them developing countries. (Data was also obtained in the same survey for average deposit size, the product of which with deposit numbers is a measure of monetary depth). However, the Beck et al. survey covers only those institutions regarded as “deposit money banks” and as such does not cover most microfinance institutions.

⁵ The developing country household survey percentages used here are those listed by Claessens (2006); the original surveys were conducted with various methodologies and at various dates between 1988 and 2004.

⁶ For example, the European Commission stresses that its Eurobarometer survey was designed to elicit attitudes about the importance of various financial issues rather than measuring usage, therefore the statistics on usage should therefore be treated with caution. These Eurobarometer surveys have been repeated on a roughly annual frequency with some startling year-to-year jumps in the responses, despite only small change in the questions answered.

⁷ Christen et al. define their coverage as including intermediaries that have a “double bottom-line”, reflecting social (access to finance) as well as strictly commercial objectives. They include government-owned development banks and savings banks. Scrutiny of the underlying institution-level data in the CGAP database revealed a handful of countries for which one or more very large institutions were counted twice; adjustments to deal with this were made to the data for: Burkina Faso, Sri Lanka, Madagascar, Niger, Senegal and Viet Nam.

⁸ A drawback here is that many of the banks added by the WSBI do not actually report the number of accounts; instead, for these banks, Peachey and Roe imputed account numbers on the assumption that the average account balance is 0.24 times annual mean per capita GDP in the country (they based the multiplier on the average ratio for countries where the relevant data exists). For 15 countries, estimated savings bank account numbers exceed 10% of the national adult population. While their simplicity and retail emphasis makes the imputation much more realistic for most savings banks than it would be for commercial banks, it is questionable for a number of the WSBI member banks which have effectively evolved into full-service commercial banks. For this reason, we have made a new examination of all WSBI members for which the Peachey and Roe imputation method generates account numbers exceeding 12% of the adult population, and made sizable downward adjustments to 7 countries (Angola, Bolivia, Cape Verde, Morocco, Malta and Mongolia) based on information contained in the annual reports of the savings banks in question and on consulting local experts on the appropriateness of the imputation.

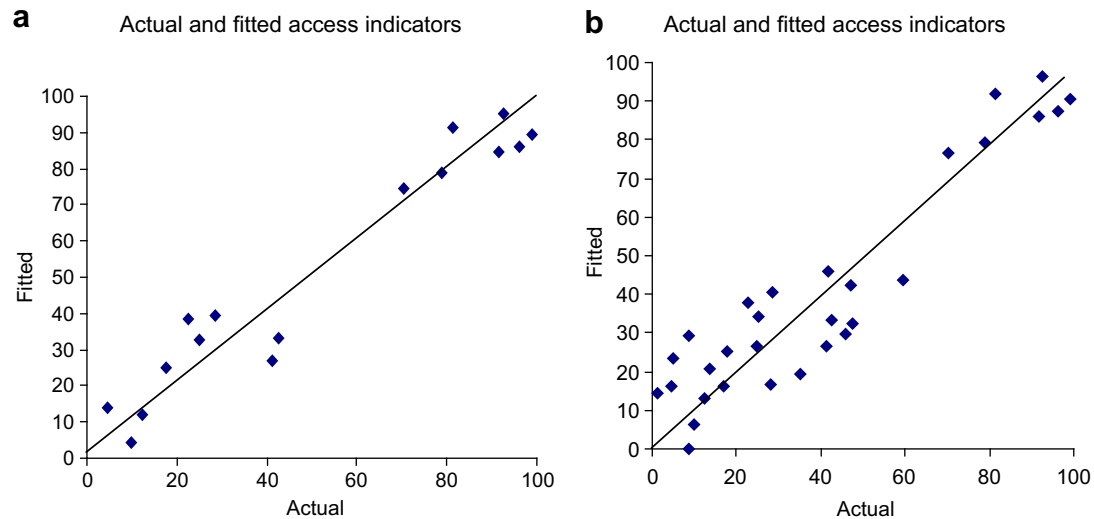


Fig. 1. Fit of the equations linking: (a) Actual and (b) synthetic bank accounts-based information with household survey-based access data (based on Regressions 1.3a and 1.3b Table 1).

A number of difficulties arises in moving from these two databases to a reliable indicator of the proportion of households with financial access.⁹ One major issue is the problem of inferring the number of distinct and active *accountholders* from data on numbers of *accounts*. For one thing, many individuals may have multiple accounts. On the other hand, some accounts are recorded even though they have long been inactive. There is sure to be a large variation from country to country and between different types of banks in the degree to which these problems result in the aggregate number of accounts exceeding the number of individuals with accounts. The typical individual in more advanced financial systems likely holds a greater number of accounts than in less advanced economies. And there are widely differing practices with regard to multiple accounts. For example, the ICBC – with the largest number of individual accounts of any bank – recently conducted a campaign to consolidate multiple accounts: for about 150 million customers, it had maintained over 400 million accounts.¹⁰

The issue of dating is also relevant. The various sources relate to different dates for different financial intermediaries. Mostly these are close to the year 2000, but there is considerable variation. For most countries it may safely be assumed that these numbers do not change dramatically within a few years; the rise of microfinance means that this is not necessarily true for all countries.¹¹

Some of the surveys are based on household units, others use the individual adult as the unit (for example the LSMS program uses the household as the unit, whereas Finscope work with the individual). The two approaches need not give the same result in terms of the proportion of the sample with an account. After all, many households may have no more than one adult with an account. As such, the proportion of households with an account is

likely to be somewhat higher than the proportion of adults with an account. Still, the difference may not be all that great. Here we simply use survey data based on both concepts interchangeably. The question of how best to splice data from the two types of sources deserves further study, but would seem to require more overlapping data than yet exists.

There are other loose ends. For instance, while the accounts at ICBC are included in the WSBI data, the 146 million active personal deposit accounts at another large Chinese institution, the China Commercial Bank are not, essentially because it is not a member of the WSBI. Both of these banks would be included in the concept of “commercial banks” employed by Beck et al. (2007b) for their survey, though data for China has not been included in that survey.

3.2. Combining different sources for a composite indicator

Despite these difficulties, we examined whether these different sources could credibly be spliced together to extend coverage to the union of all of the countries covered in any of the datasets.

Thus, we define for each country i :

b_i = the number of bank accounts per 100 adults (Beck et al., 2007b series);

h_i = household survey-based percentage access (as described above);

m_i = the number of accounts at MFIs and alternative intermediaries per 100 adults (WSBI series as modified);

y_i = per capita GDP;

z_i = average deposit size;

We fit the following OLS regressions, reported in the upper panel of Table 1

$$b_i = \alpha_0 + \alpha_1 \log(m_i) + u_i, \quad (1)$$

$$z_i = \beta_0 + \beta_1 y_i + u_i, \quad (2)$$

where α_j , β_j , are estimated parameters (as is γ_j below), and u_i is an identically and independently distributed disturbance.

Although the number of available observations is small, the fit in both cases is encouraging. The number of bank accounts, and average bank deposit size, can be plausibly approximated by the fitted values of Regressions 1.1 and 1.2 respectively.

⁹ For example, there is unfortunately an overlap between the coverage of the two databases on numbers of accounts, as some deposit money banks are included in the WSBI sample. Rather than attempting to eliminate the overlaps, we have – as explained in the text – in effect chosen to aggregate the two sources in a way that is not inconsistent with the existence of overlaps.

¹⁰ The bank's notice to customers dated October 18, 2005, urging them to close or merge accounts containing small balances, identifies no fewer than eleven classes of account which should *not* be closed. The figure of 150 million customers comes from the bank's initial public offering (IPO) prospectus, released in October 2006.

¹¹ For the regression results reported in Section 3, the other explanatory variables (and the poverty and inequality variables) refer, for the most part, to 2000 or the late 1990s.

Table 2

Composite measure of access to financial services

Albania	b	ALB	34	Georgia		GEO	15	Pakistan	b	PAK	12
Algeria		DZA	31	Germany	s	DEU	97	Panama		PAN	46
Angola		AGO	25	Ghana		GHA	16	Papua New Guinea	b	PNG	8
Antigua & Barbuda		ATG	48	Greece		GRC	83	Paraguay		PRY	30
Argentina	b	ARG	28	Grenada		GRD	37	Peru		PER	26
Armenia	s	ARM	9	Guatemala		GTM	32	Philippines		PHL	26
Austria	s	AUT	96	Guinea		GIN	15	Poland	s	POL	66
Azerbaijan		AZE	17	Guyana	s	GUY	20	Portugal	s	PRT	84
Bahamas, The		BHS	53	Haiti		HTI	15	Romania	s	ROM	23
Bangladesh		BGD	32	Honduras	b	HND	25	Russian Federation		RUS	69
Barbados		BRB	56	Hong Kong, Chn		HKG	[38]	Rwanda		RWA	23
Belarus		BLR	16	Hungary	s	HUN	66	Samoa		WSM	19
Belgium	s	BEL	97	India	s	IND	48	Sao Tome & Principe		STP	15
Belize		BLZ	46	Indonesia		IDN	40	Saudi Arabia	b	SAU	62
Benin		BEN	32	Iran, Isl. Rep.		IRN	31	Senegal		SEN	27
Bermuda		BMU	48	Iraq		IRQ	17	Seychelles		SYC	41
Bhutan		BTN	16	Ireland	s	IRL	88	Sierra Leone		SLE	13
Bolivia		BOL	30	Italy	s	ITA	75	Singapore	b	SGP	98
Bosnia & Herzeg.		BIH	17	Jamaica	s	JAM	59	Slovak Republic		SVK	83
Botswana	s	BWA	47	Jordan	b	JOR	37	Slovenia		SVN	97
Brazil	s	BRA	43	Kazakhstan		KAZ	48	Solomon Islands		SLB	15
Bulgaria	s	BGR	56	Kenya	s	KEN	10	South Africa	s	ZAF	46
Burkina Faso		BFA	26	Korea, Rep.		KOR	63	Spain	s	ESP	95
Burundi		BDI	17	Kyrgyz Republic	s	KGZ	1	Sri Lanka		LKA	59
Cambodia		KHM	20	Latvia	s	LVA	64	St. Kitts and Nevis		KNA	49
Cameroon		CMR	24	Lebanon	b	LBN	[79]	St. Lucia		LCA	40
Canada	s	CAN	96	Lesotho	s	LSO	17	St. Vincent & Gren.		VCT	45
Cape Verde		CPV	40	Liberia		LBR	11	Sudan		SDN	15
Cent African Rep.		CAF	19	Libya		LYB	27	Suriname		SUR	32
Chile		CHL	60	Lithuania	s	LTU	70	Swaziland	s	SWZ	35
China	s	CHN	42	Luxembourg	s	LUX	99	Sweden	s	SWE	99
Colombia	s	COL	41	Macao, China		MAC	[14]	Switzerland	b	CHE	88
Comoros		COM	20	Macedonia, FYR		MKD	20	Syrian A.R.		SYR	17
Congo, Rep.		COG	27	Madagascar		MDG	21	Tajikistan		TJK	16
Costa Rica		CRI	29	Malawi		MWI	21	Tanzania	s	TZA	5
Cote d'Ivoire	s	CIV	25	Malaysia	b	MYS	60	Thailand		THA	59
Croatia		HRV	42	Mali		MLI	22	Timor-Leste		TMP	13
Cuba		CUB	45	Malta	s	MLT	90	Togo		TGO	28
Cyprus	s	CYP	85	Mauritania		MRT	16	Trinidad & Tobago		TTO	53
Czech Republic	s	CZE	85	Mauritius	b	MUS	54	Tunisia		TUN	42
Denmark	s	DNK	99	Mexico	s	MEX	25	Turkey	b	TUR	49
Dominica		DMA	66	Moldova		MDA	13	Uganda		UGA	20
Dominican Rep.		DOM	29	Mongolia		MNG	25	Ukraine		UKR	24
Ecuador	s	ECU	35	Morocco		MAR	39	United Kingdom	s	GBR	91
Egypt		EGY	41	Mozambique		MOZ	12	United States	s	USA	91
El Salvador		SLV	26	Myanmar		MMR	19	Uruguay		URY	42
Eritrea		ERI	12	Namibia	s	NAM	28	Uzbekistan		UZB	16
Estonia	s	EST	86	Nepal		NPL	20	Venezuela	b	VEN	28
Ethiopia		ETH	14	Netherlands		NLD	100	Vietnam		VNM	29
Fiji	b	FJI	39	Nicaragua	s	NIC	5	West Bank & Gaza		WBG	14
Finland	s	FIN	99	Niger		NER	31	Yemen, Rep.		YEM	14
France	s	FRA	96	Nigeria		NGA	15	Yugoslavia, FR		YUG	21
Gabon		GAB	39	Norway	b	NOR	84	Zambia		ZMB	15
Gambia		GMB	21	Oman		OMN	33	Zimbabwe	b	ZWE	34

Note: 's' means household survey data used; 'b' means fitted data using bank deposit numbers and not WSBI numbers. [Square bracketed data are considered less reliable – see footnote 15]. This is Mark III version of the data series: as more refined information on specific countries is obtained, it is envisaged that this series will be updated. Source: Author's calculations based on Beck et al. (2007b), Christen et al. (2004), Claessens (2006), European Commission (2005), Peachey and Roe (2006) and subsequent revisions. For definitions and method see Section 3.

Using these fitted values $\hat{b}_i$ and $\hat{z}_i$ for countries where b_i and z_i are not available,¹² we then proceed to estimate:

$$h_i = \gamma_0 + \gamma_1 \log \hat{b}_i + \gamma_2 \log \hat{z}_i + u_i. \quad (3)$$

Once again, this regression of household survey-based data on access percentages on bank account numbers and average bank account size provides a close enough fit to suggest using extrapolating fitted values for countries in which there is no

household survey data available (Table 1, Regressions 1.3 and Fig. 1).^{13,14}

These coefficient values can be used to approximate access percentages for all 138 countries in the WSBI database. Our proposed composite indicator of household access uses survey data h_i for countries where it is available, and the fitted values from (1.3a) otherwise. This series is presented in Table 2, and in Figs. 2 and 3.

¹² I.e. $\hat{b}_i = b_i$ for countries where data on b_i is available; $\hat{b}_i = \hat{\alpha}_0 + \hat{\alpha}_1 \log(m_i)$, the fitted value from (1), otherwise; $\hat{z}_i = z_i$ for countries where data on z_i is available; $\hat{z}_i = \hat{\beta}_0 + \hat{\beta}_1 y_i$, the fitted value from (2) otherwise.

¹³ Including the date of the household survey in the regressions changes little. Several other functional forms were tried: those reported are the best-fitting.

¹⁴ Regression 1.3b in Table 1 reports the same regression estimated over a restricted sample using only observations for which b_i and z_i are available; it is almost identical.

To recapitulate and summarize, our main new data series, the composite indicator, is constructed as follows. For any country with data on access from a household survey, this surveyed percentage is copied in directly. For other countries the percentage is constructed as a function of the estimated number, and average size of, bank accounts. Thus, if there is no survey data on bank account numbers, an estimate of bank account numbers is generated as a function of MFI account numbers; if there is no data on average deposit size, an estimate of average deposit size is generated as a function of GDP per capita. The estimated access percentage is then generated as a function of the estimated number of bank accounts and average deposit size. Each of the functional relationships used is based on an estimated regression relationship fitted to the countries for which the relevant data are available.

Despite the evident shortcomings of the composite access indicator, it has the great merit of distilling information from a variety of different sources and placing it on a common basis. No doubt the accuracy of the imputation for any given country may be questionable, but it represents a first approximation to be revised as new data become available for each country. Furthermore, it certainly has potential for use in cross-country regression analysis, where econometric techniques can minimize the impact of stochastic errors.

3.3. Features of the composite indicator

Looking at how the access indicator varies across regions of the developing world, Latin America and the Caribbean have the highest mean and median percentages, but the variation within each region is quite considerable. The lowest mean and median are for Africa and for the developing countries of Eastern Europe and Central Asia.

Fig. 2 plots the most widely used measure of the scale of financial intermediation through the banking system, namely private credit as a percentage of GDP (which we call banking depth for short) against the access series. The correlation between the two series is low ($R = 0.32$), making it clear that access and depth represent quite different dimensions of the financial sector.

Fig. 3 shows a strong and positive correlation between national mean per capita income and access percentages, but also reveals a wide variation in estimated access percentages across countries at similar levels of mean income.¹⁵

Digging deeper, we have regressed the composite access indicator on a set of country structural characteristics in order to throw light on the question: what kinds of country tend to have more access? The cross-sectional regressions we estimated included, in addition to GNI per capita (measured at purchasing power parities), demographic variables (age dependency, birth rate and size of population), the structure of production (share of agriculture in GNI), geography (population density per square km), aid dependency (aid received as a percentage of GNI), technology (mobile phone subscribers per 1000 persons) and institutions (illiteracy percentage and the average of the KKZ governance indicators, originally introduced by Kaufmann et al., 1999).

All of these variables appear significantly in some of the regressions, a representative selection of which are reported in Table 3. Interestingly, however, GNI per capital fails to be significant in the regressions that exclude high income countries (such as

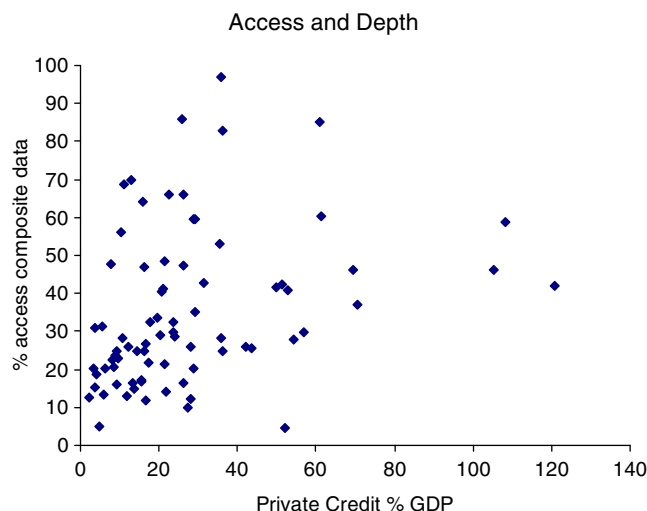


Fig. 2. Access and banking depth (credit to private sector as % GDP). Access is the financial access indicator of Table 2. Credit data (1999) from International Financial Statistics.

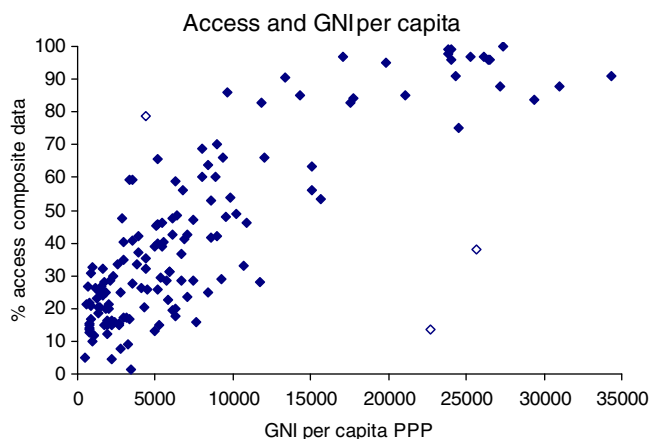


Fig. 3. Access and per capita income – 144 countries. Access is the financial access indicator of Table 2. GNI per capita data (1999) from World Development Indicators. Note: the two outliers on the lower right hand side are Hong Kong and Macao, China; that at the top left hand side is Lebanon.

Regressions 3.1 and 3.2).¹⁶ Evidently, other correlates of income, such as the indicator of institutional governance indicator and the penetration of mobile phones, are more reliably associated with financial access. Indeed, the robust and highly significant coefficient on the mobile phone variable is very striking. The share of agricultural production is negatively associated with financial access, possibly reflecting the physical remoteness of farmers and farm workers from financial service providers. While overall density of population is negatively associated with access, this counterintuitive relationship vanishes when the two largest outliers, Hong Kong and Lebanon, are excluded (Regressions 3.3 and 3.4).¹⁷ The most significant of the demographic variables is age dependency, which is negatively associated with access. An African dummy, often significant in other contexts, is positively related to access, but fails to reach statistical significance.

¹⁶ The aid received variable is not defined for donor countries and therefore regressions that include the aid variable exclude *ipso facto* the high income countries.

¹⁷ The sample for Regression 3.2 already excludes Lebanon because of missing explanatory data.

¹⁵ The widest outliers are two low estimated percentages for Hong Kong and Macao; these reflect the low WSBI figures for these countries, and do not use bank account data, even though at least for Hong Kong we know the banking system to be highly developed and likely generating higher outreach than estimated. On the other hand, the figure for Lebanon is well above the scatter data reflecting the use of its banking data even though its banking sector is exceptionally deep reflecting its role as a regional banking centre.

Table 3
Financial access and country characteristics

Equation	3.1		3.2		3.3		3.4	
	Coefficient	t-Stat	Coefficient	t-Stat	Coefficient	t-Stat	Coefficient	t-Stat
Constant	75.6	2.1	46.5	**7.6	44.2	**7.8	10.2	0.0
GNI per cap (log)	−0.859	0.2					5.27	2.2
Share of agriculture in GNI	−0.274	1.7						
Aid as % GNI	−0.528	2.0	−0.635	**3.3	−0.605	**3.4		
Age dependency (%)	−0.527	2.2	−0.171	*2.0	−0.161	*2.1	−0.167	*1.7
Birth rate	0.251	0.7						
Population (log)	−1.14	1.2						
Density of population 000/km ²	−4.85	2.4	−5.70	**3.0	−1.64	0.1		
Mobile phone subscribers per 000	0.060	**3.7	0.071	**5.0	0.076	**5.7	0.055	**6.1
Institutions (KKZ index)	4.40	1.3	9.62	*4.0	8.54	**3.8	8.10	**3.3
Illiteracy	0.093	0.9						
Africa dummy	4.76	1.2						
Outliers omitted?	No		No		HKG, LBN		HKG, LBN	
R-squared/NOBS	0.641	103	0.648	128	0.699	127	0.820	139
Adjusted R-squared	0.597		0.634		0.686		0.814	
S.E. of regression	12.5		12.6		11.6		11.9	

This table shows a representative sample of OLS regressions relating financial access to income, productive structures, aid, demography, geography, technology and institutions. ** and * indicates significance at the 1% and 5% levels, respectively.

Dependent variable is the financial access series of Table 2.

Source for explanatory variables: KKZ data for 1998 from Kaufmann et al. (1999); others from *World Development Indicators*. (GNI: 1999; Illiteracy: 2001; others 2000).

The Africa dummy takes the value 1 for all sub-Saharan African countries except South Africa, zero otherwise. Cross section: all available except Hong Kong and Lebanon omitted in Regressions 3.3–3.4. (The residuals for these two countries in regression 3.2 are −33.1 and 32.2, respectively).

4. Financial access and poverty: Is there a cross-country link?

While recognizing the shortcomings of the imputation procedures used to construct the composite indicator, as well as the limitations (outlined in the previous section) of the underlying data sources, the resulting dataset nevertheless seem worth using in econometric exercises to throw light on the question of whether financial access reduces poverty.

Earlier work (using the CGAP data also described above) failed to find any significant cross-country correlation between the density of microfinance accounts and poverty rates (Honohan, 2004b). But given the extensive differences between the old data and the new composite series, it is worth repeating the experiment.

In fact, as shown in Regression 4.1 (Table 4), access is correlated with the \$1 per day poverty headcount rate.¹⁸ But access percentages are, as already shown by Fig. 3, strongly correlated with per capita income, and since the latter is highly correlated with poverty rates, the more interesting question is whether access remains significant in a regression which also includes per capita income. In fact, it does not (Regression 4.2).¹⁹

Regressions 4.3–4.5 illustrates the results of adding four categories of control found to be significant in earlier work were added. First, the depth of private banking credit (expressed as a percentage of GDP) and its companion, the inflation rate;²⁰ second, measures of political and social institutional quality; third, population size; and fourth, a dummy for countries in sub-Saharan Africa. Collinearity between these variables and the presence of several wide outliers mean that little beyond average income is significant when the full sample is used (Regressions 4a–g), though the Africa dummy

is significant and positive when the institutional and population size variables are excluded.

Removing outliers (the note to regression 4.2 shows how wide these outliers are) reveals some more structure. In particular, even with average income included, banking depth tends to be significant, with the sign implying that deeper systems have less poverty (Regressions 4.4–4.5).²¹ This is consistent with the results reported in Honohan (2004a,b), despite the inclusion of the financial access variable, which remains insignificant. In other words, if financial development lowers poverty, it is in its depth dimension rather than the access dimension that this is evident in cross-country data.²²

In the absence of plausible instruments, all of the reported regressions have used ordinary least squares rather than an instrumental variables estimator. This is not likely to be as serious a problem when the dependent variable is poverty count or inequality, as it would be if the equations were trying to explain income levels and growth.²³

5. Concluding remarks

For low-income countries, the relevant question for poor households is not how much financial assets they have, but whether they have access to financial intermediaries at all. We have drawn on and synthesized recent work by several authors using information provided by banks and micro-finance institutions in combination with household survey data to produce estimates of access percentages for some 160 countries. These estimates can be progressively refined as the results of a new wave of access surveys

¹⁸ This is the proportion of the population consuming less than \$1.08 in 1993 purchasing power parity equivalent. It is widely used as a simple measure of absolute poverty, cf. Ravallion and Chen (2001).

¹⁹ Following Honohan (2004a), regression 4.2 includes the mean income of the bottom 90%, together with the percentage share of the top decile; including just mean income likewise eliminates the significance of access (not shown).

²⁰ Since all variables related to monetary depth are highly correlated across country with inflation (because agents economize on money balances when inflation is high), it is prudent to include inflation as a control variable if monetary depth is being used as a structural measure of financial development (Honohan, 2004a).

²¹ The institutional variables, when significant, tend to have a counterintuitive sign – better institutions meaning more poverty – but these coefficients are not robust.

²² A similar exercise was conducted with the \$2 per day poverty headcount as the dependent variable; the same pattern holds (regression estimates not reproduced here): financial access is correlated with the broader poverty definition but not when the effects of average income is controlled-for.

²³ Of course there could still be an endogeneity problem, for example, if both poverty and access were driven by a common unobserved factor. But what is absent here is the kind of automatic feedback from average income to financial depth that makes instrumenting essential in the causal analysis of finance and growth, for example.

Table 4
Poverty and financial access

Equation	4.1		4.2		4.3		4.4		4.5	
	Coefficient	t-Stat	Coefficient	t-Stat	Coefficient	t-Stat	Coefficient	t-Stat	Coefficient	t-Stat
Constant	65.6	**6.1	139.9	**8.4	130.3	**4.1	158.5	**6.1	160.2	**11.3
GNI per cap lower 90% (log)			–18.3	**8.5	–14.3	**3.4	–17.4	**5.6	–19.4	**9.4
Share of top 10%			0.555	**3.1	0.428	2.0	0.307	2.0	0.431	**3.0
Access (log)	–13.7	**4.3	1.5	0.6	–0.50	0.1	3.32	0.9	4.27	1.3
Private credit (log % of GDP)					–0.72	0.2	–5.18	2.1	–5.26	3.0
Inflation (log)					–0.34	0.2	–0.76	0.5		
Institutions (KKZ index)					1.96	0.4	5.09	1.4		
Institutions (Freedom House bank)					–3.52	1.3	–2.02	1.1		
Africa dummy					11.00	1.8	7.30	1.6		
Outliers omitted?	No		No		No		Yes (see note)		Yes (see note)	
R-squared/ NOBS	0.175	89	0.644	87	0.685	69	0.802	64	0.788	66
Adjusted R-squared	0.166		0.631		0.643		0.773		0.774	
S.E. of regression	20.7		13.7		13.9		9.6		9.9	

This table shows a representative sample of OLS regressions relating the percentage of the population below the \$1 per day poverty line to financial access percentages across countries. ** and * indicates significance at the 1% and 5% levels, respectively.

Dependent variable is the percentage of the population below the \$1 per day poverty line.

Access is the financial access series of Table 2.

Source for poverty and income distribution data (from last available survey before 2004), GNI (1999, measured at PPP), Inflation (average of CPI inflation 1990–2000), and all other explanatory variables: *World Development Indicators*, except Private Credit (1999) *International Financial Statistics*; KKZ data for 1998 from Kaufmann et al. (1999).

The Africa dummy takes the value 1 for all sub-Saharan African countries except South Africa, zero otherwise.

Cross section: Regressions 4.1–4.2: all available; Regressions 4.3–4.5 exclude the six countries with the largest residuals from equation 4.2, namely Ethiopia (residual = 27.4); Mongolia (–21.7); Nicaragua (43.1); Tanzania (–29.3); Uganda (50.0); Yemen (–25.4).

currently under way or planned become available. They should prove useful to researchers and policymakers alike.

Interestingly, higher mobile phone penetration and better institutions of governance are correlated across countries with the new access series, even after controlling for per capita income.

The new access percentages are negatively correlated across countries with poverty headcount rates, whether at the \$1 a day or \$2 a day thresholds. But the correlation is not a robust one and loses significance in multiple regressions that include mean per capita income. Thus the supposed anti-poverty potential of financial access remains econometrically elusive.²⁴

References

- Armendariz de Aghion, B., Morduch, J., 2005. *The Economics of Microfinance*. MIT Press, Cambridge, MA.
- Beck, T., Demirgüç-Kunt, A., Levine, R., 2007a. Finance, inequality and poverty: Cross country evidence. *Journal of Economic Growth* 12 (1), 211–252.
- Beck, T., Demirgüç-Kunt, A., Martínez Peria, M.S., 2007b. Reaching out: Access to and use of banking services across countries. *Journal of Financial Economics* 85 (1), 234–266.
- Carbó, S., Gardener, E.P.M., Molyneux, P., 2005. *Financial Exclusion*. Palgrave Macmillan, Basingstoke.
- Christen, R.P., Jayadeva, V., Rosenberg, R., 2004. Financial institutions with a double bottom line: Implications for the future of microfinance. Consultative Group to Assist the Poor Occasional Paper No. 8.
- Claessens, S., 2006. Access to financial services: A review of the issues and public policy objectives. *World Bank Research Observer* 21, 207–240.
- Demirgüç-Kunt, A., Beck, T., Honohan, P., 2008. *Finance for All: Policies and Pitfalls in Expanding Access*. The World Bank, Washington, DC.
- European Commission, 2005. “Public Opinion in Europe on Financial Services.” Special Eurobarometer 230. Luxembourg. <http://europa.eu.int/comm/consumers/cons_int/fina_serv/cons_experiences/background_en.htm>.
- Honohan, P., 2004a. Financial development, growth and poverty: How close are the links? In: Goodhart, C. (Ed.), *Financial Development and Economic Growth: Explaining the Links*. Palgrave, London.
- Honohan, P., 2004b. *Financial Sector Policy and the Poor: Selected Issues and Evidence*. The World Bank, Washington DC.
- Honohan, P., 2005. Measuring microfinance access: Building on existing cross-country data. World Bank Policy Research Working Paper No. 3606.
- Honohan, P., 2008. Household financial assets in the process of development. In: Davies, J.B. (Ed.), *Personal Wealth from a Global Perspective*. Oxford University Press, Oxford.
- Kaufmann, D., Kraay, A., Zoido-Lobaton, P., 1999. Aggregating governance indicators. World Bank Policy Research Working Paper No. 2195.
- Kempson, E., Whyley, C., Caskey, J., Collard, S., 2000. In or Out? Financial Exclusion: A Literature and Research Review. Working Paper. Financial Services Authority, London.
- Peachey, S., Roe, A., 2006. Access to finance, measuring the contribution of savings banks. Working Paper. World Savings Bank Institute.
- Ravallion, M., Chen, S., 2001. Measuring pro-poor growth. World Bank Policy Research Working Paper No. 2666.
- Robinson, M.S., 2001. *The Microfinance Revolution: Sustainable Finance for the Poor*. The World Bank, Washington, DC.

²⁴ A corollary, that the major identified influence of financial development on poverty reduction seems to come through its growth effect mediated through the business sector, is explored in some detail in Demirgüç-Kunt et al., 2008.